

Hirdianshu Singla

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Education

BASc, Computer Engineering, *The University of British Columbia, Vancouver*

Expected Graduation: Dec 2027

Relevant Coursework:

Algorithms & Theory: DSA; Algorithms (design & analysis); Probability & Statistics

Systems & Hardware: Operating Systems; Computer Architecture; Digital Systems/FPGA

Machine Learning & AI: (Intelligent Systems – search, CSPs, planning, reasoning under uncertainty); (ML &

Data Mining – regression, classification, clustering, neural networks, gradient descent)

Work Experience

Trainee Analyst

Apr 2024 – Aug 2024; Jun 2025 – Sep 2025

Winux Software Solutions Pvt Ltd, *Mohali, Punjab*

- Led 8+ discovery sessions with ~10 stakeholders and authored 30+ user stories, shifting defect discovery one sprint earlier
- Mapped as-is/to-be flows (BPMN/draw.io), consolidated feedback, and maintained a live change log to keep development and QA aligned
- Collaborated with QA to derive test scenarios and edge cases, improving test coverage and earlier defect detection
- Implemented Python scripts to parse logs, compute pass/fail statistics, and generate first-fail triage summaries

Technical Projects

CHESS AI Platform

Jan 2026 – April 2026

- Trained a feedforward neural network on 50,000 Stockfish-evaluated positions using engineered 70-dimensional board feature vectors to predict position quality
- Designed and deployed a minimax engine with alpha-beta pruning; integrated ML model as an alternative evaluator with live comparison against engine output
- Built real-time two-player multiplayer via Socket.io with room-based matchmaking, move validation, and disconnect handling across a full TypeScript/Express REST API

F1TENTH Autonomous Racing,

Sept 2025 – Dec 2025

- Implemented a motion-control stack using MPCC for overtaking on a 1/10-scale car, integrating LiDAR and vision inputs for perception and path tracking
- Developed LiDAR-based obstacle detection and vision processing pipeline to feed real-time environment data into the motion planner
- Collaborated on TTC-based safety braking and telemetry logging to support reproducible tuning and consistent deployment across test tracks

FPGA Baccarat

Jan 2026 – Feb 2026

- Designed a full baccarat game engine in SystemVerilog using an FSM-driven dealing pipeline with correct third-card rules and mod-10 hand scoring. Implemented an LFSR-based pseudorandom card rank generator.
- Authored a self-checking simulation testbench verifying FSM state transitions, result encoding, and edge cases; analyzed waveforms to confirm correct behaviour; synthesized and deployed on DE1-SoC FPGA hardware

Technical Skills:

Programming: Java; Python; C/C++; Object-Oriented Programming; Data Structures & Algorithms

Machine Learning & Data: PyTorch; neural network design & training; feature engineering; model evaluation (R^2 , loss curves); Stockfish/engine integration

Software Engineering & Tools: Requirements analysis; Git; Linux; Bash/Make; debugging with GDB; unit testing

Hardware, FPGA: Verilog; SystemVerilog (assertions, testbenches, FSM & FIFO design); ModelSim/Quarta; Quartus; DE1-SoC FPGA board.

Robotics: ROS2; LiDAR sensing & obstacle detection; motion planning (MPCC); autonomous vehicle control; TTC-based safety systems; real-time sensor fusion.

